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Cleft lip and palate management

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outline

- Introduction
- Embryology
- Etiology
- Classification
- Deformities
- Rx planning and timing
- Cleft lip and palate repair

Introduction

- ❖ The dictionary meaning of cleft is a crack, fissure, split or a gap.
- ❖ The orofacial clefts are congenital deformities, which manifest at birth.
- ❖ Any disturbance during the embryological formation and development and growth of orofacial region will result in the formation of orofacial clefts.
- ❖ Orofacial clefts are the most frequently occurring craniofacial birth defects.

Cont.

- The management of these deformities is challenging and requires multidisciplinary approach, complex long term treatment plan and a rehabilitation programme designed for the individual case.

Incidence

- Clefts of the upper lip and palate are the most common major congenital craniofacial abnormality
- Present in approximately 1 in 700 live births
- Clefts of the lip occur more commonly in males than in females.
- Left-sided cleft lips are more common than right-sided cleft lips
- Unilateral cleft lips are more common than the bilateral cleft of the lip.
- Bilateral clefts of the lip are most often associated with clefting of both the primary and secondary palates.

Cont.

- Cleft palate alone is seen in approximately 1 in 2,000 live births and this incidence is similar in all racial groups.
- Significant differences in the prevalence of clefts exist when specific ethnic/ racial populations are examined.
- For example, African Americans have a birth prevalence that is less common than the total population, but Asians tend to have a higher prevalence.

Cont.

- In the majority of cases unilateral cleft lip and palate is an isolated nonsyndromic birth defect that is not associated with any other major anomalies.
- By comparison a much greater proportion of patients with an isolated cleft palate will have an associated syndrome or sequence.
- Some of the more common syndromes seen in this group include Stickler's, Van der Woude's, or DiGeorge syndromes.

Embryology

- At approximately 6 weeks of human embryologic development the median nasal prominence fuses with the lateral nasal prominences and maxillary prominences to form the base of the nose, nostrils, and upper lip.
- The confluence of these anterior components becomes the primary palate.
- When this mechanism fails, clefts of the lips and/or maxilla occur.

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- At approximately 8 weeks the palatal shelves elevate and fuse with the septum to form the intact secondary palate.
- When one palatal shelf fails to fuse with the other components, then a unilateral cleft of the secondary palate occurs.
- If both of the palatal shelves fail to fuse with each other and the midline septum, then a bilateral cleft of the palate occurs.

Cont.

- Fusion occurs when programmed cell death (apoptosis) occurs at the edges of the palatal shelves.
- The ectodermal component disintegrates and the mesenchyme fuses to form the intact palate.
- Soon after this the anterior primary palate fuses with the secondary palate and ossification occurs.

Cont.

- At any point, if failure of fusion occurs with any of the above components, a cleft will occur of the primary and/or secondary palates.
- Clefts may be complete or incomplete based on the degree of this failure of fusion.
- Currently ultrasound images of clefts of the lip can be visualized as early as 16 weeks.

Genetics and etiology

- Clefts are thought to be of a multifactorial etiology with a number of potential contributing factors.
- These factors may include chemical exposures, radiation, maternal hypoxia, teratogenic drugs, nutritional deficiencies, physical obstruction, or genetic influences.
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Cont.

- Recently multiple genes have been implicated in the etiology of clefting.
- Some of these genes include the FOXE1, IRF6, MSX, LHX, goosecoid, and DLX genes.

Cont.

- Disturbances in growth factors or their receptors may be involved in the failure of fusion .
- ❖ This include fibroblast growth factor, transforming growth factor, platelet-derived growth factor, and epidermal growth factor.
- The chances of a recurrence of clefting within a family are dependent on many factors, including family history, severity, gender, degree of relationship to the affected individual, and the expression of a syndrome.

classification

- ✓ The typical classification system used clinically to describe standard clefts of the lip and palate is based on careful anatomic description.
- ✓ Clefts can be unilateral or bilateral; microform, incomplete, or complete; and may involve the lip, nose, primary palate, and/or secondary palate.
- ✓ Veau
- ✓ Kernahan

veau

Veau (1931) presented a very simple classification

- Group I: Cleft of the soft palate only
- Group II: Cleft of hard and soft palate
- Group III: Complete unilateral cleft, extending from uvula to incisive foramen and then deviates to one side extending through the alveolus.
- Group IV: Complete bilateral alveolar cleft.



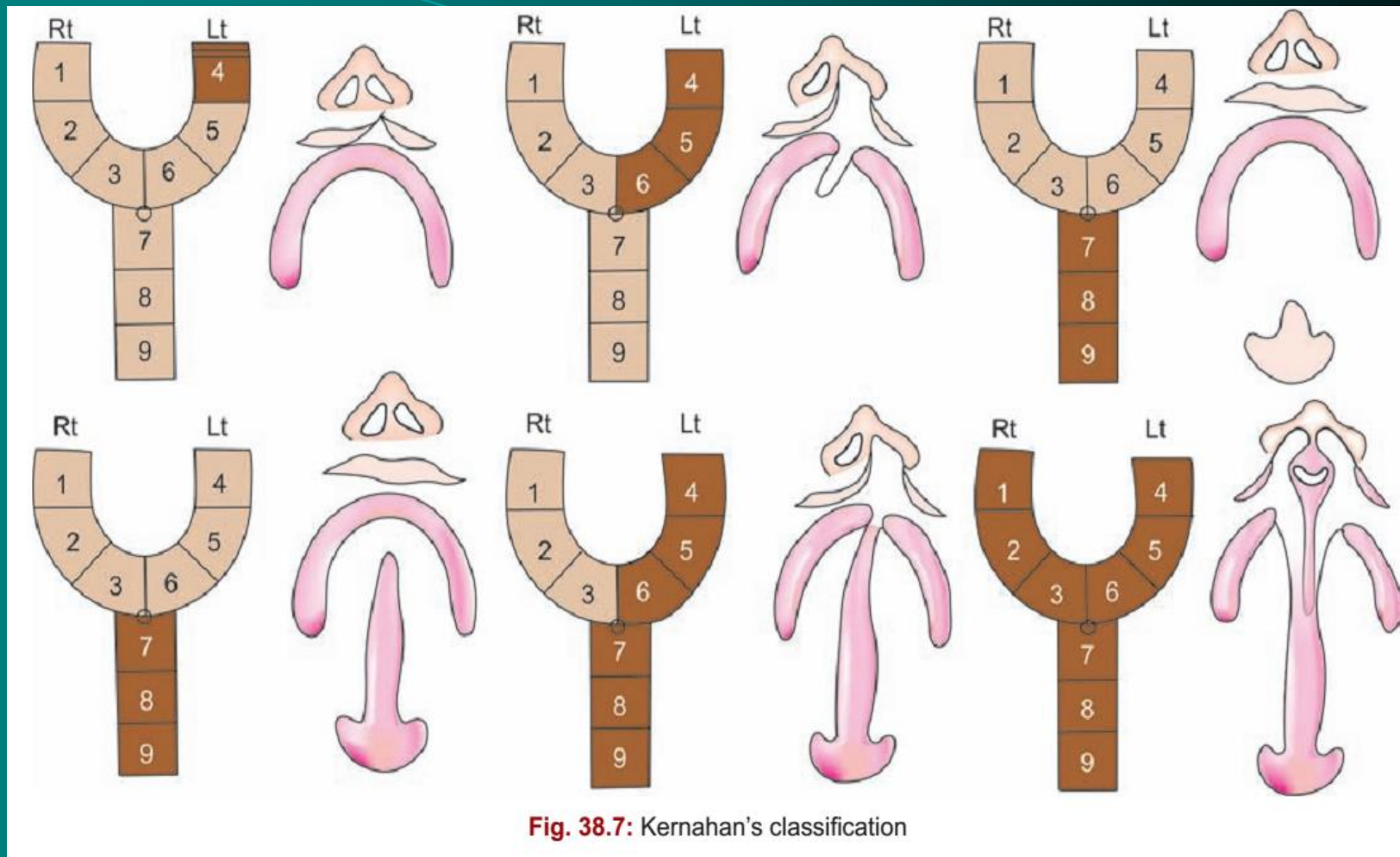
FIGURE 43-1. Cleft lips come in a variety of configurations, such that each repair must be customized to establish the most normal morphology. **A**, Microform left unilateral cleft lip only, not requiring primary repair. **B**, Minor left incomplete unilateral cleft lip only. **C**, Left incomplete unilateral cleft lip and palate with a Simonart's band. **D**, Wide left complete unilateral cleft lip and palate.



Figs 38.1A to C: Bilateral complete cleft lip and cleft palate deformity. (Courtesy: Dr Amresh S Baliar Singh, Head, Department of Plastic Surgery, KEM Hospital, Mumbai, Maharashtra, India)

kernahan

- ❖ Kernahan has simplified it by representing various clefts in the form of a Y.
- ❖ The anterior portions of the Y depict the lip (1 and 4), the middle alveolus (2 and 5), incisive foramina and the posterior portion (the area of the hard palate from the alveolus back to the incisive foramen (3 and 6), posterior to the incisive foramen, the hard (7 and 8) and the soft (9) palate.



Deformities and complications

- ❖ Feeding
- ❖ Speech
- ❖ Hearing
- ❖ Dental
- ❖ Growth disturbances

Feeding

- Infants with cleft palate can have difficulty feeding due to the inability to form an adequate seal between the tongue and palate for creation of sufficient negative pressure to suck fluid from a bottle.
- Nasal regurgitation and inefficient handling of secretions and foodstuffs may also be observed during early development.
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- In infants that have a wide cleft of the secondary palate, where breast-feeding may not be possible a combined protocol that includes intermittent feeding with the use of a specialized bottle and attempts at nursing(breastfeeding) is the preferred approach.
- How and importance?

Cont.

- ✓ Specialized nipples and bottles are necessary to improve feeding immediately after birth.
- ✓ The problem can be overcome through the use of specially designed nipples, that have elongated and bigger opening, which extend directly into the hypopharynx.
- ✓ Spoon feeding also helps.
- ✓ During feeding the child will swallow lot of air, therefore the child should not be fed while recumbent and more frequent burping is necessary.

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- ✓ Feeding sessions should last no longer than 35 minutes.
- ✓ Close attention to weight gain is necessary for these children.
- ✓ Generally, in 24 hours each infant should have approximately 2 to 3 ounces of milk for each pound of weight.
- ✓ Infants should be weighed at least weekly using the same scale.

Dental

- ❖ Cleft of the alveolus may affect the development of the primary and permanent teeth and the jaw itself.
- ❖ Teeth can be congenitally absent or supernumerary teeth may be seen or crowding or displacement of permanent teeth can be seen.
- ❖ Usually the cleft is seen extending between the lateral incisor and canine areas.
- ❖ The teeth may be morphologically deformed or hypomineralized and the eruption can be seen at the cleft margin.
- ❖ The hypoplastic maxilla will have common finding of pseudoprognathism or relative prognathism of the mandible.

Speech

- ❖ Retardation of consonant sounds is the most common problem (p, b, t, d, k and g).
- ❖ Speech is affected, hypernasality can remain in these patients, even after the surgical correction.
- ❖ Dental malformation and abnormal tongue position may produce an articulation problem.
- ❖ Hearing impairment aggravates the speech problem further.
- ❖ The cleft palate patient will be handicapped with multiple problems of hearing loss from middle ear infection.

Cont.

- ✓ The growth of the maxilla will be affected in these patients resulting into functional and esthetics impairment and malocclusion.
- ✓ In Pierre Robin syndrome along with cleft palate, there is mandibular micrognathia and glossoptosis.
- ✓ The underdeveloped mandible results in tongue fall in the oropharynx, airway obstruction in newborn child.

Treatment planning and timing

- ❖ The timing of cleft lip and palate repair is controversial.
- ❖ Understanding the growth and development of the craniofacial skeleton is critical to the treatment planning process.
- ❖ In many cases waiting for a greater degree of growth to occur is advantageous unless compelling functional or esthetic issues are present that can not or should not wait.

Cont.

- ❖ Cleft lip repair is generally undertaken at some point after 10 weeks of age.
- ✓ Allows a complete medical evaluation of the patient so that any associated congenital defects affecting other organ systems (eg, cardiac or renal anomalies) may be uncovered.
- ✓ The surgical procedure itself may be easier when the child is slightly larger and the anatomic landmarks more prominent and well defined.
- ✓ Anesthesia related risk

Cont.

- Historically the anesthetic risk-related data suggested that the safest time period for surgery in this population of infants could be outlined simply by using the “rule of 10’s
- This referred to the idea of delaying lip repair until the child was at least 10 weeks old, 10 pounds in weight, and with a minimum hemoglobin value of 10 dL/mg.

Cont.

- ❖ Cleft palate repair is usually performed at approximately 9 to 18 months of age.
- ❖ Approximately 75% of patients with any type of cleft will present with clefting of the maxilla and alveolus.
- ❖ Bone graft reconstruction of this site is performed during the mixed dentition prior to the eruption of the permanent canine and/or the permanent lateral incisor.
- ❖ The timing of this procedure is based on dental development and not chronologic age

Cont.

- ❖ Orthognathic reconstruction of maxillary and mandibular discrepancies is performed at 14 to 18 years of age based on individual growth characteristics.
- ❖ This is done in conjunction with orthodontics prior to and after surgery.
- ❖ As with the timing of other interventions, lip and nasal revision is best reserved until after the majority of growth is complete.
- ❖ Lip revision can be considered prior to school age at about 5 years of age

TABLE 43-1. Staged Reconstruction of Cleft Lip and Palate Deformities

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Procedure	Timing
Cleft lip repair	After 10 weeks
Cleft palate repair	9–18 months
Pharyngeal flap or pharyngoplasty	3–5 years or later based on speech development
Maxillary/alveolar reconstruction with bone grafting	6–9 years based on dental development
Cleft orthognathic surgery	14–16 years in girls, 16–18 years in boys
Cleft rhinoplasty	After age 5 years but preferably at skeletal maturity; after orthognathic surgery when possible
Cleft lip revision	Anytime once initial remodeling and scar maturation is complete but best performed after age 5 years

Cleft lip and palate repair

- **Presurgical Taping and Presurgical Orthopedics**
- **Surgical**

Reference

- Peterson principles of oral and maxillofacial surgery 4th edition
- Neelima Anil Malik, Textbook of Oral & Maxillofacial Surgery- 4th edition :Jaypee
- Peterson principles of oral and maxillofacial surgery 2nd edition